Sound sources of different levels

- Suppose we have two loudspeakers, the first playing a sound with power P1, and another playing a louder version of the same sound with power P2, but everything else (how far away, frequency) kept the same.
- The difference in decibels between the two is given by $10 \log (P2/P1)$ dB
- If the second produces twice as much power than the first, the difference in dB is $10 \log (P2/P1) = 10 \log 2 = 3$ dB.

Adding SPLs of independent sound sources

- Since SPLs are based on a log scale, they cannot be added directly
  - I.e., 80 dB + 80 dB $\neq$ 160 dB

$$\boxed{SPL_T = 10 \times \mathrm{Log}\left(\sum_{i=1}^{n} 10^{\left(\frac{SPL_i}{10}\right)}\right)}$$

  - Where: SPL_T is the total sound pressure level, and SPL_i is the ith sound pressure level to be summed

- Given two machines producing 80 dB each, what is the total SPL?

$$SPL_T = 10 \times \mathrm{Log}\left(\sum_{i=1}^{n} 10^{\left(\frac{SPL_i}{10}\right)}\right)$$

$$= 10 \times \mathrm{Log}\left(10^{(80/10)} + 10^{(80/10)}\right)$$

$$= 10 \times \mathrm{Log}\left(2 \times 10^{8}\right)$$

$$= 83\, dB$$

- Important rule of thumb …
- Adding two sound pressure levels of equal value will always result in a 3 dB increase!
  - 80 dB + 80 dB = 83 dB
  - 100 dB + 100 dB = 103 dB
  - 40 dB + 40 dB = 43 dB

- Given four machines producing 100 dB, 91dB, 90 dB, and 89 dB respectively, what is the total sound pressure level?

$$SPL_T = 10 \times \mathrm{Log}\left(\sum_{i=1}^{n} 10^{\left(\frac{SPL_i}{10}\right)}\right)$$

$$= 10 \times \mathrm{Log}\left(10^{(100/10)} + 10^{(91/10)} + 10^{(90/10)} + 10^{(89/10)}\right)$$

$$= 10 \times \mathrm{Log}\left(10^{10} + 10^{9.1} + 10^{9} + 10^{8.9}\right)$$

$$= 101.2 dB$$

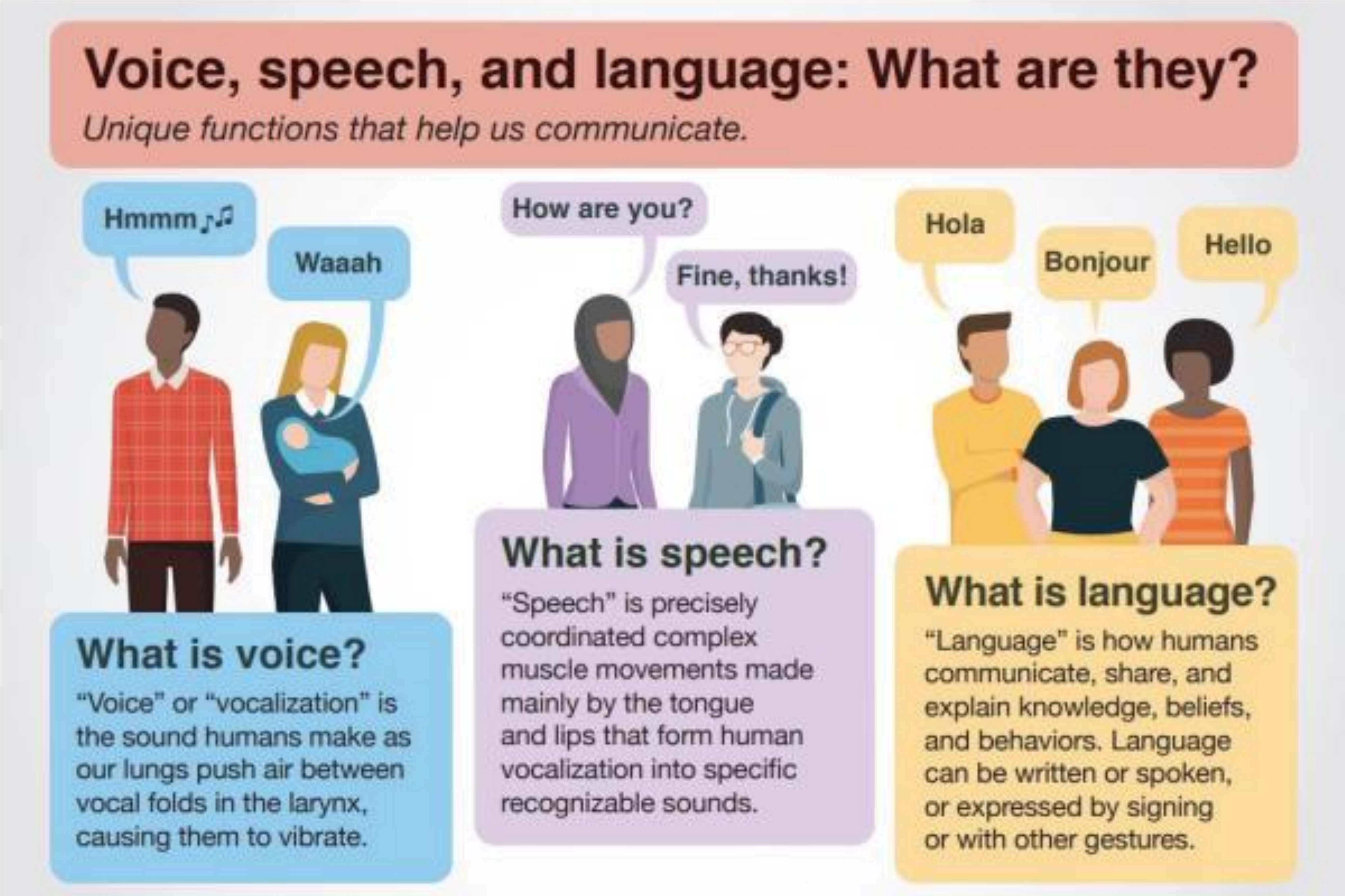
Adding Decibels Rule-of-Thumb C(ΔL)

- If you are adding two dB values, L_{high} and $L_{low,}$
- for $\Delta L > 9$, $L_{sum} = L_{high}$
- for $\Delta L = 4...9$, $L_{sum} = L_{high} + 1$
- for $\Delta L = 2...3$, $L_{sum} = L_{high} + 2$
- for $\Delta L = 0...1$, $L_{sum} = L_{high} + 3$
- Ex: 55 + 65 =65

  60 + 65 =66

  65 + 65 + 65 = 68 + 65 = 70

Speech production and acoustic properties


Voice, speech, and language: What are they?
Unique functions that help us communicate.
Hmmm ♪♫
Waaah
How are you?
Fine, thanks!
Hola
Bonjour
Hello
What is voice?
"Voice" or "vocalization" is the sound humans make as our lungs push air between vocal folds in the larynx, causing them to vibrate.
What is speech?
"Speech" is precisely coordinated complex muscle movements made mainly by the tongue and lips that form human vocalization into specific recognizable sounds.
What is language?
"Language" is how humans communicate, share, and explain knowledge, beliefs, and behaviors. Language can be written or spoken, or expressed by signing or with other gestures.

Physiological speech production

- Oscillations are primarily produced when the *vocal folds (vocal cords)* are tensioned appropriately.
- This produces *voiced sounds* and is perhaps the most characteristic property of speech signals.
- Vowel sounds (A, E, I, O, U) and diphthongs (combinations of two vowel sounds) are all voiced. That also includes the letter Y when pronounced like a long E.(Examples: city, pity, gritty.)
- Sounds without oscillations in the vocal folds are known as *unvoiced sounds*.

Voiced and Unvoiced

Voiced

/b/ /d/ /g/ /j/ /l/
/m/ /n/ /r/ /v/
/w/ /y/ /z/

/th/ that

All vowels
a e i o u

Vibration in vocal chords
NO push of air

Unvoiced

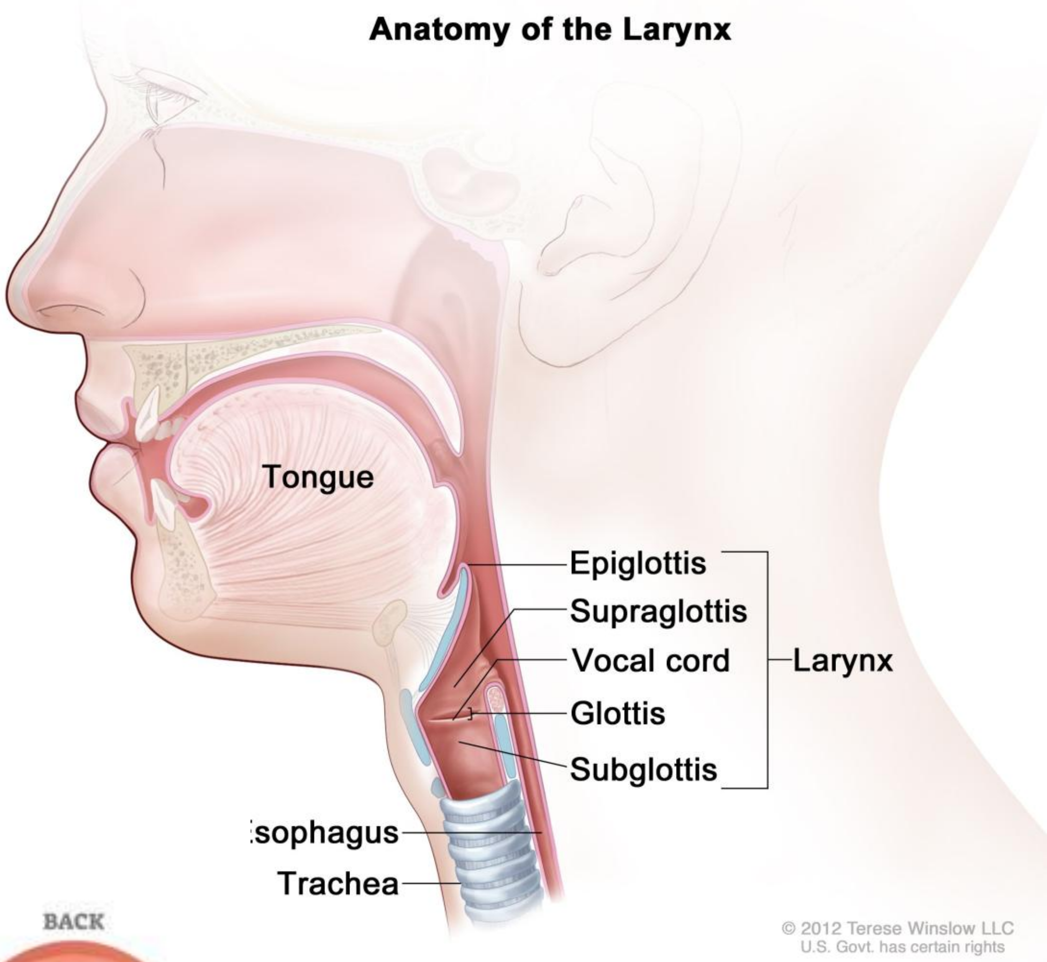
/f/ /h/ /k/ /p/
/s/ /t/ /x/
/qu/ /ch/ /sh/
/th/ thing

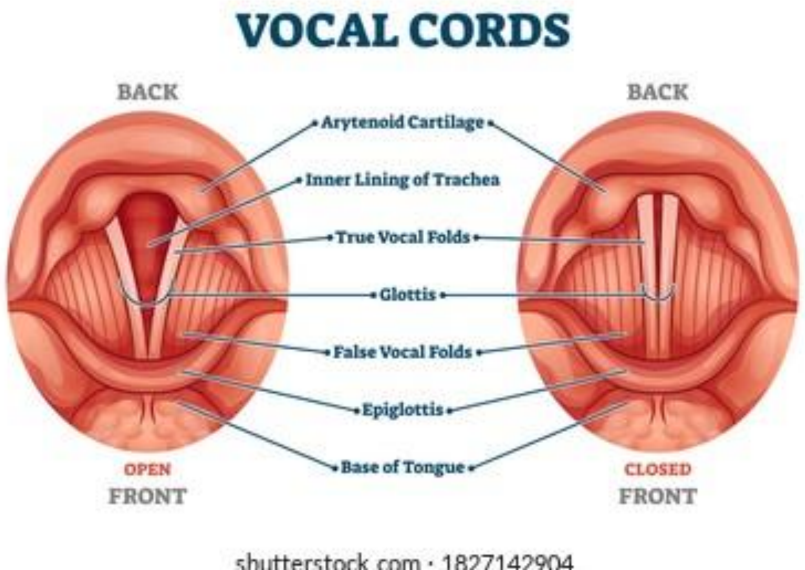
NO vibration in vocal chords
Push of air

Feel with hand on throat

The vocal folds

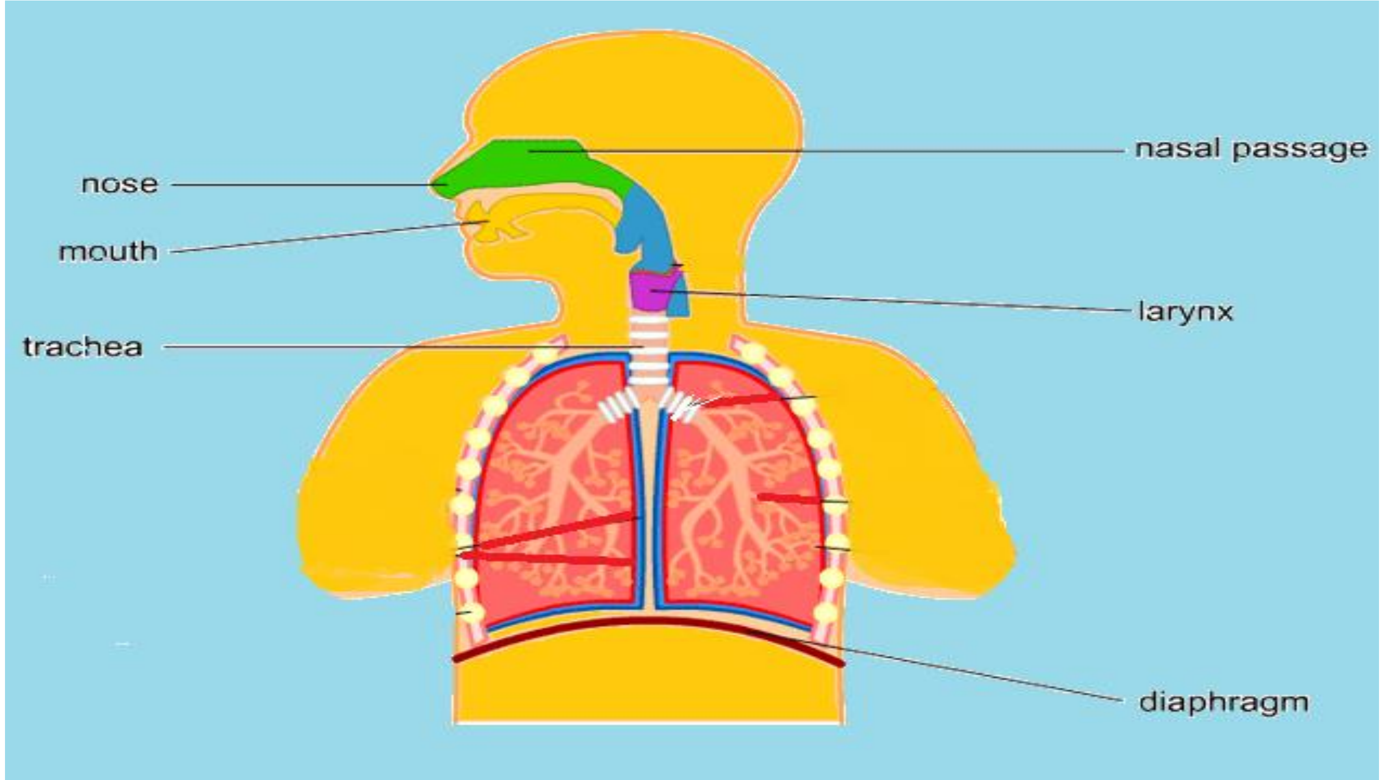
- The *vocal folds,* also known as vocal cords, are located in the throat and oscillate to produce voiced sounds.
- The opening between the vocal folds (the empty space between the vocal folds) is known as the *glottis*.
- The sudden stop of airflow is the largest acoustic event in the vocal folds and is known as the *glottal excitation*.





The production of sounds

- There are 3 processes associated with the production of speech sounds:
  - Initiation: The creation of the airstream
  - Phonation: Airstream modification by the vocal folds (cords)
  - Articulation: Further modification of airstream by articulators (lips, tongue etc)

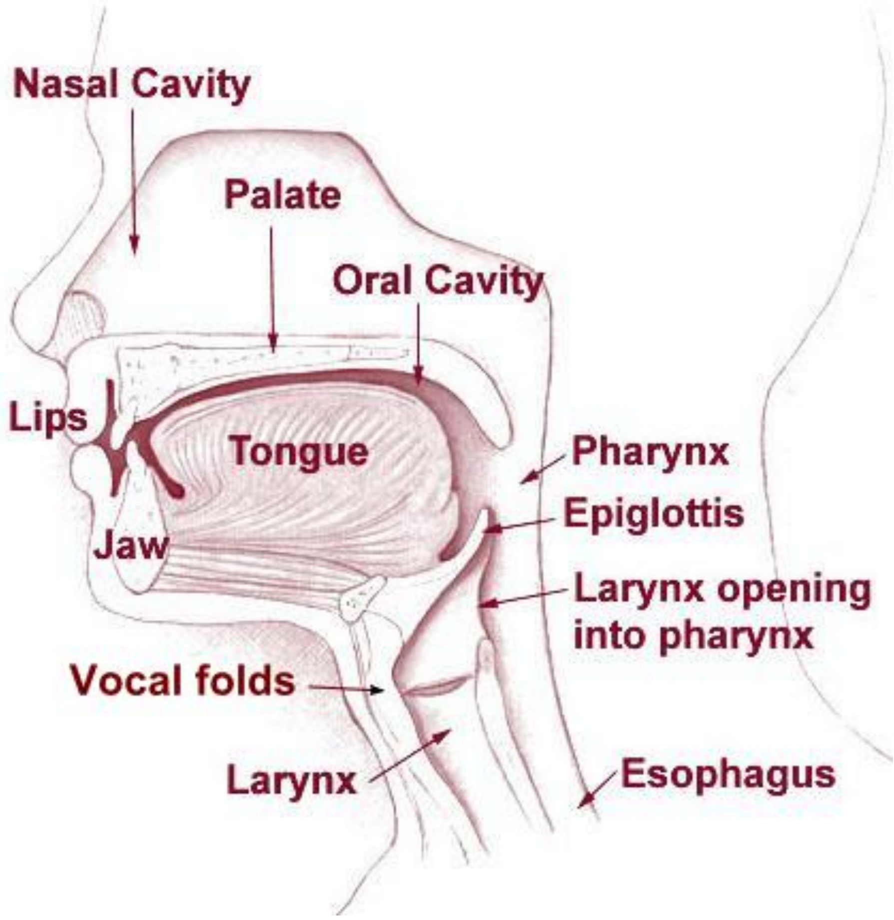


Pitch

- The frequency of vocal folds oscillation depends on:
  - Amount of lengthwise tension in the vocal folds
  - Pressure differential above and below the vocal folds
  - Length and mass of the vocal folds.
- Frequency of the vocal folds refers to the actual physical phenomenon
- *Pitch* refers to the perception of frequency.

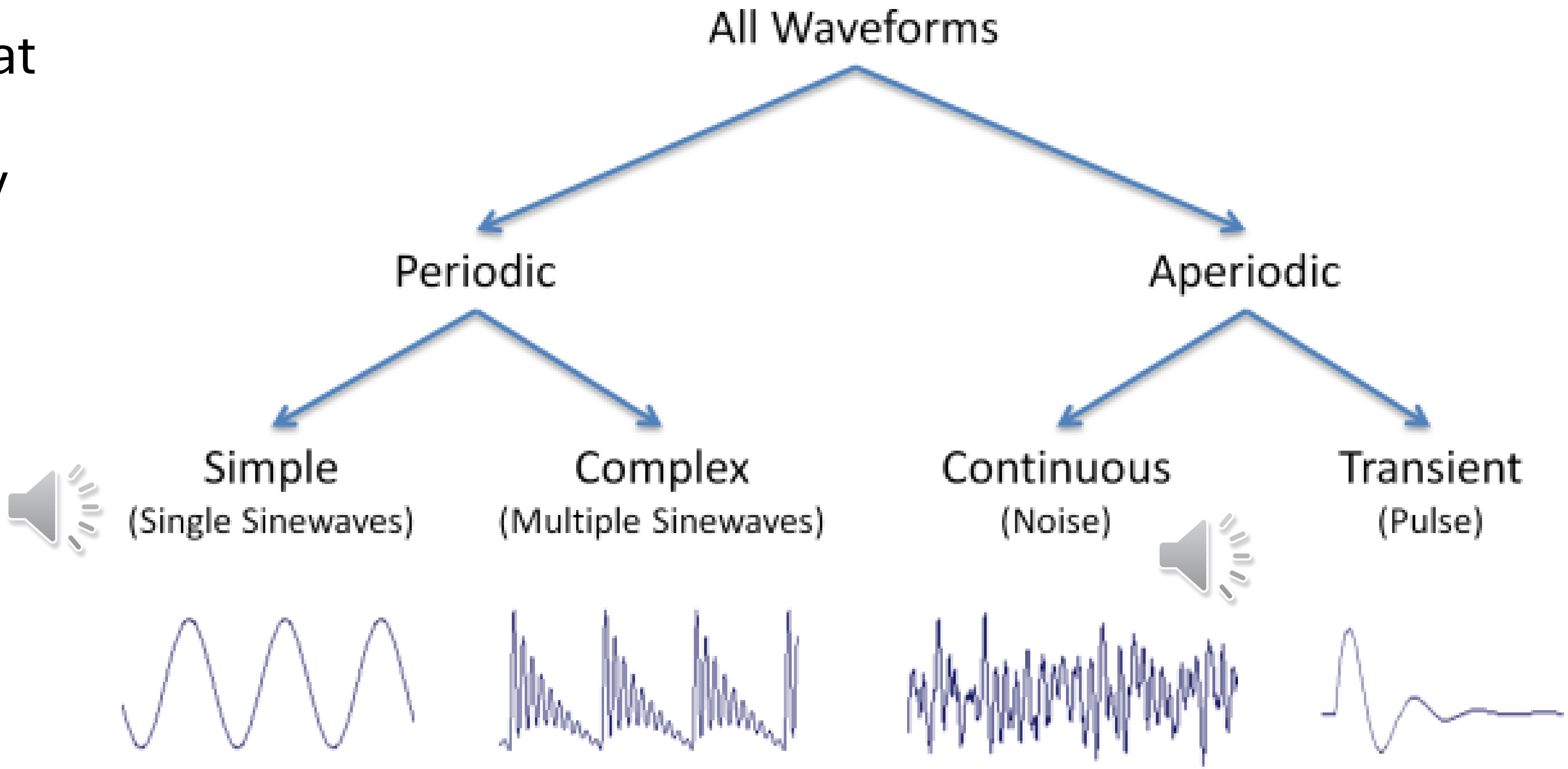
The vocal tract

- The vocal tract, including the larynx, pharynx and oral cavities, have a great effect on the pitch of the sound. Namely, the shape of the vocal tract determines the resonances and anti-resonances of the acoustic space, which boost and attenuate different frequencies of the sound.
- Specifically, the acoustic features which differentiate *vowels* from each other are the frequencies of the resonances in the vocal tract, corresponding to specific *places* of articulation primarily in terms of tongue position. Since the air can flow relatively unobstructed, vowel sounds tend to have high energy and loudness compared to *consonants*.

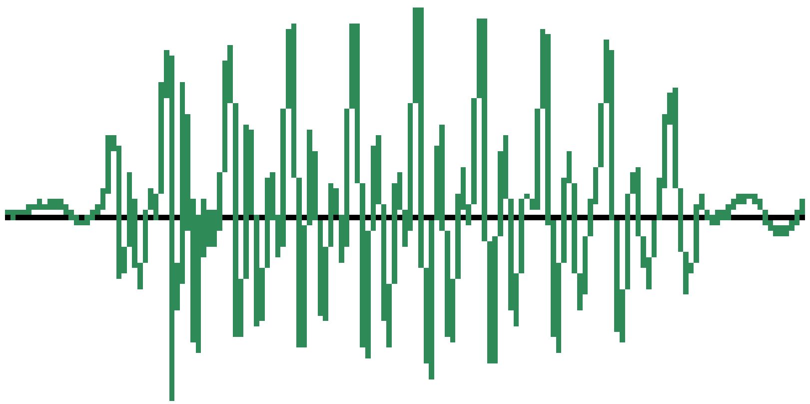


Types of sound waves
Periodic vs. aperiodic waves

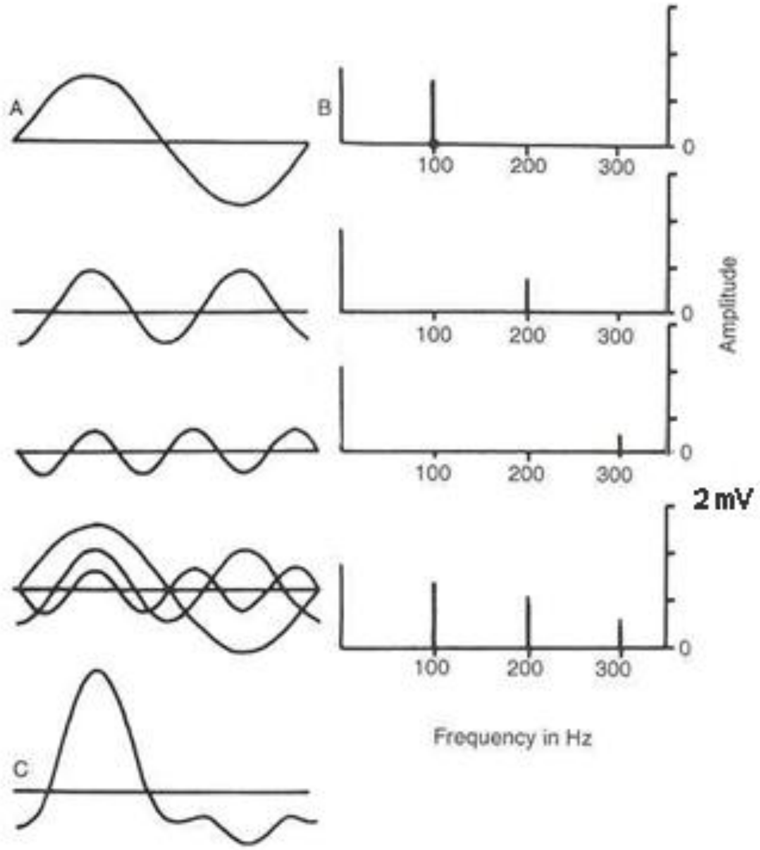
- Periodic waves repeat at regular intervals (by contrast to aperiodic ones)



- **Simple waves** are a good way to learn about basic properties of frequency, amplitude, and phase.
- Examples include whistling; not really found much in speech
- **Complex periodic waves**
- Results from imperfectly oscillating bodies
- Examples - a vibrating string, the vocal folds
- Consists of a fundamental (F_0) and harmonics
- Harmonics ("overtones") consist of energy at integer multiples of the fundamental (x2, x3, x4 etc…)

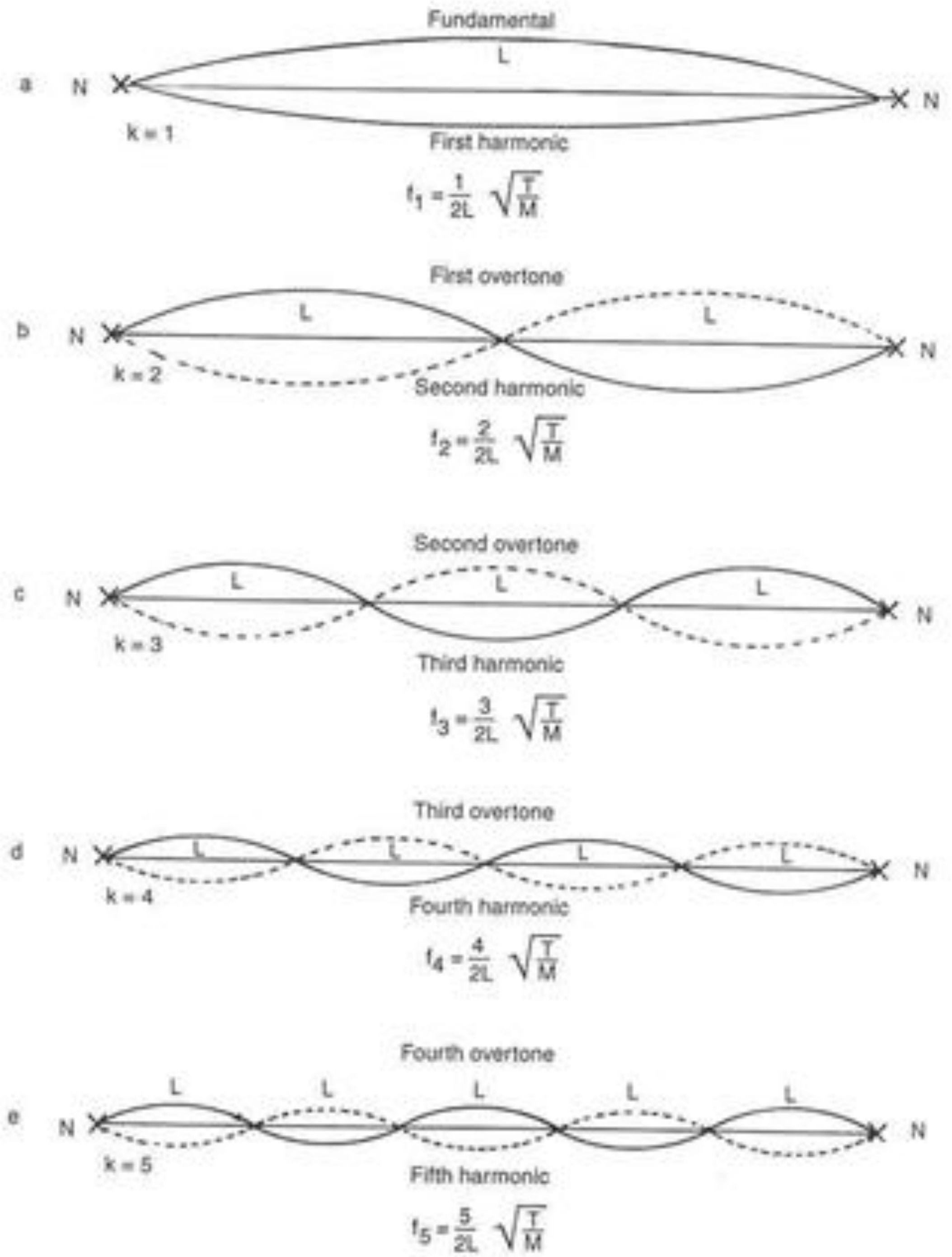


From complex wave to its components… and the frequency spectrum



(complex wave)

- Also known as a "line spectrum"
- Here, complex wave at the bottom…
- ..is broken into its component sin waves shown at the top



Harmonic series

- Imagine you pluck a guitar string and could look at it with a really precise strobe light
- Here is what its vibration will look like

Resonance of vocal tract

- *Resonant* means literally to sound again. This implies that there is a sound to resound. In voice this sound is phonation created at the larynx through the oscillating vocal folds
- Anything that can vibrate, vibrates at *natural frequencies*; when the energies of an initial sound match the object's natural frequencies, the object's material vibrates in sympathy.
- In addition, if the object contains a cavity of air, that air will vibrate at frequencies based on its size and shape. This air filled cavity is known as an *acoustic resonator*.


TACOMA BRIDGE
COLLAPSE
BRITISH PATHÉ
PATHE GAZETTE

Pitch versus volume

Phonation is a complex tone, it is made up of a *fundamental frequency* or F_o and *harmonic multiples* of the F_o ($2F_o$, $3F_o$, $4F_o$ etc.) that fall in *intensity* (volume) in an inverse relationship as the harmonics rise in frequency (pitch); or as the pitch rises the volume falls.

Acoustic properties of speech signal

- The most important acoustic features of a speech signal are
- *fundamental frequency*
- *resonance of the vocal tract*
- Signal *amplitude* or *intensity*

Physical models of speech?

Vocal-tract Model with Flexible Tongue

Takayuki Arai
(Sophia University)

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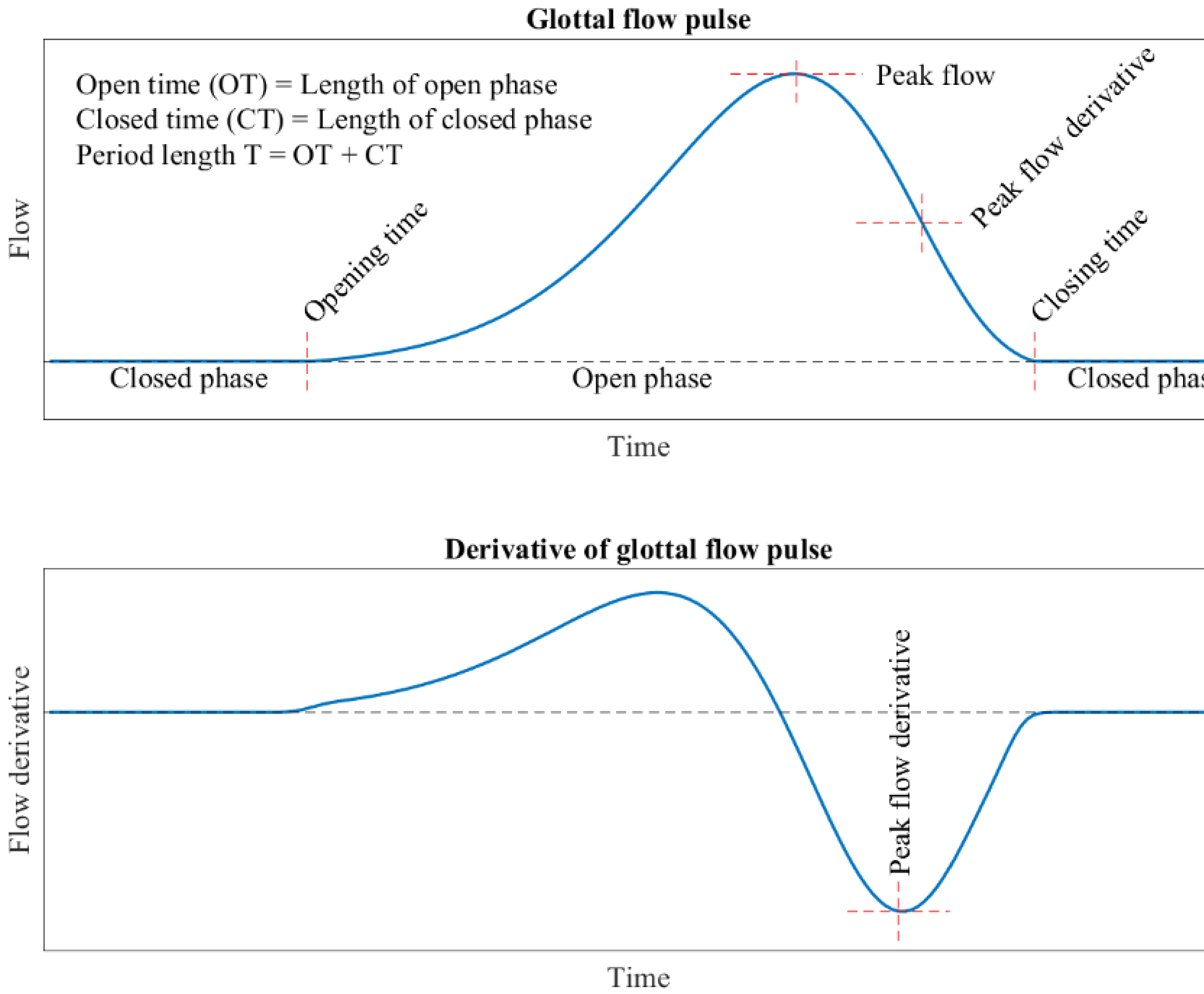
Building physical models for speech is really hard because biomechanics is hard! It's actually way easier to generate connected speech in a computer…

Glottal activity

- "agressiveness" of the pulse relates to the tension of glottal folds and is characterized by the (negative) *peak of the glottal flow derivative*.
- All parameters describing a length in time are often further normalized by the period length t.



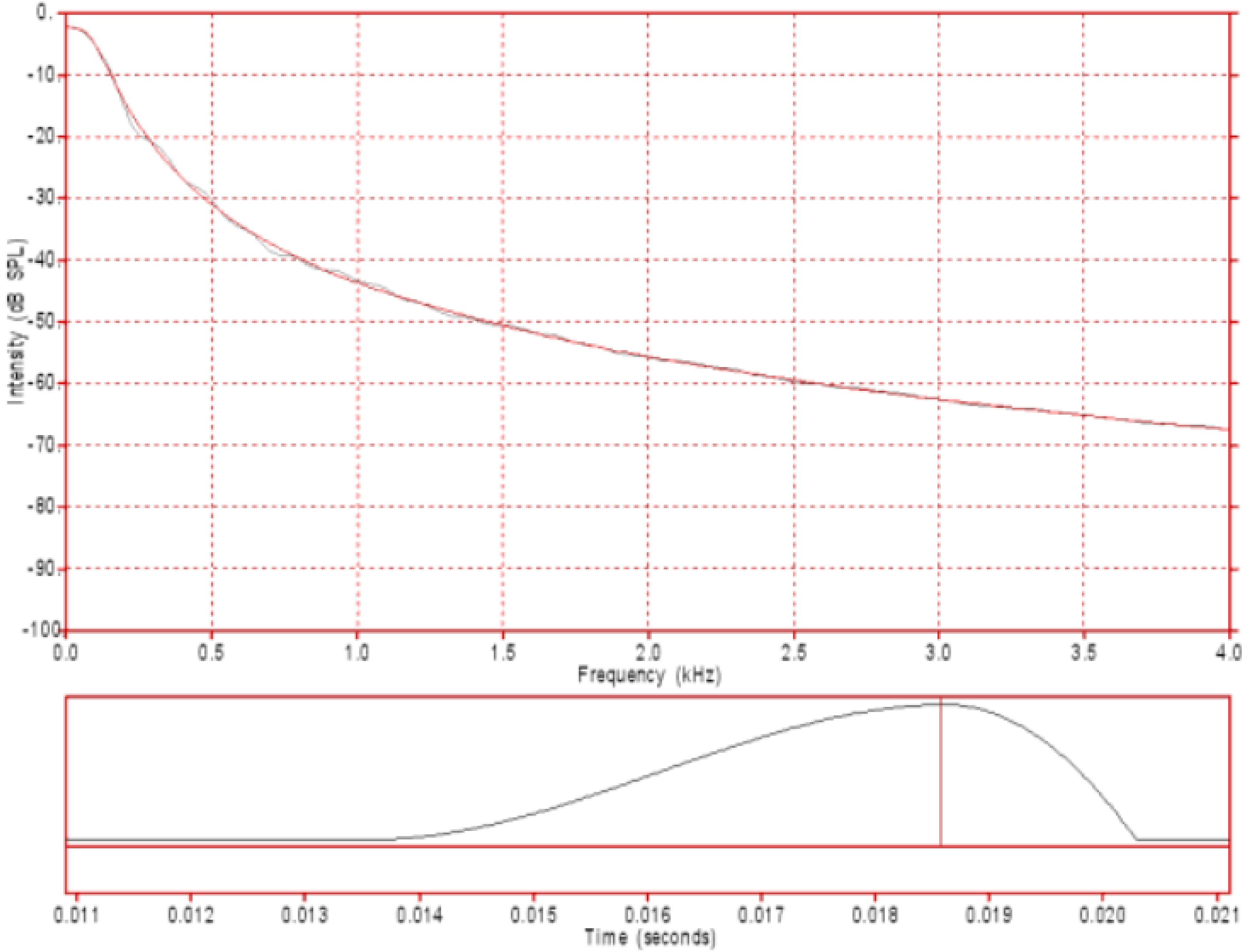


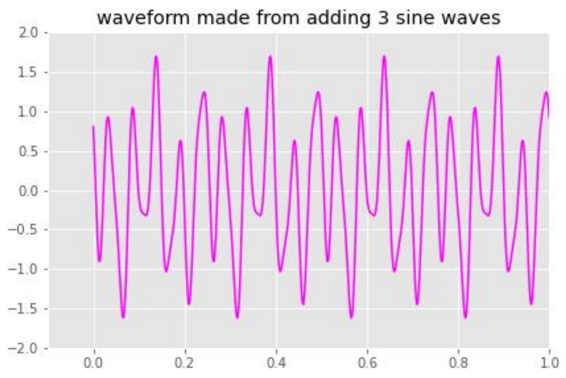
Figure 1: A single synthesised glottal pulse waveform (bottom) and its spectrum (top). This glottal pulse has a rise (opening phase) time of about 5.0 ms and a fall (closing phase) time of about 1.7 ms (opening to closing ratio of about 3:1).

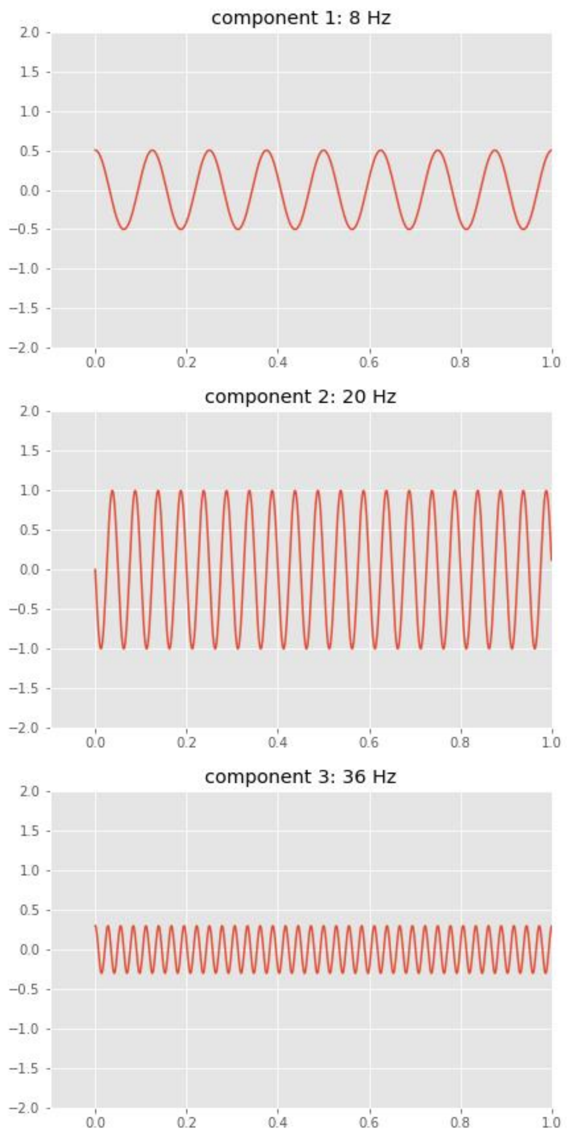
Fourier Transform

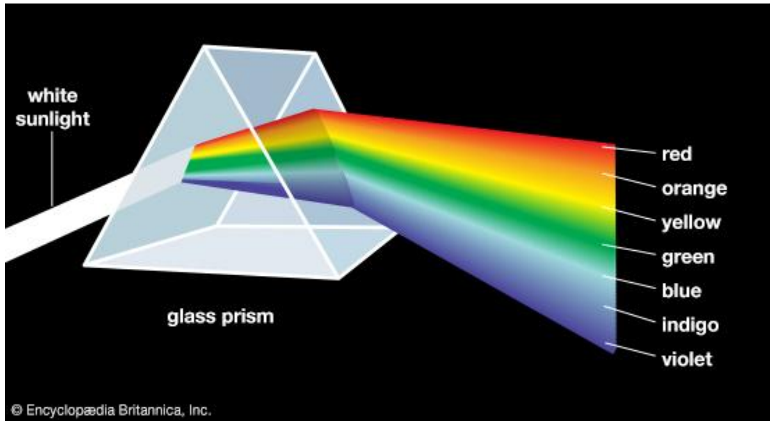
We can **decompose** a periodic waveform into a set of **simple periodic waves** (i.e., pure tones) of different frequencies.

Just like a prism splits light into component colours

A spectrum of colours!

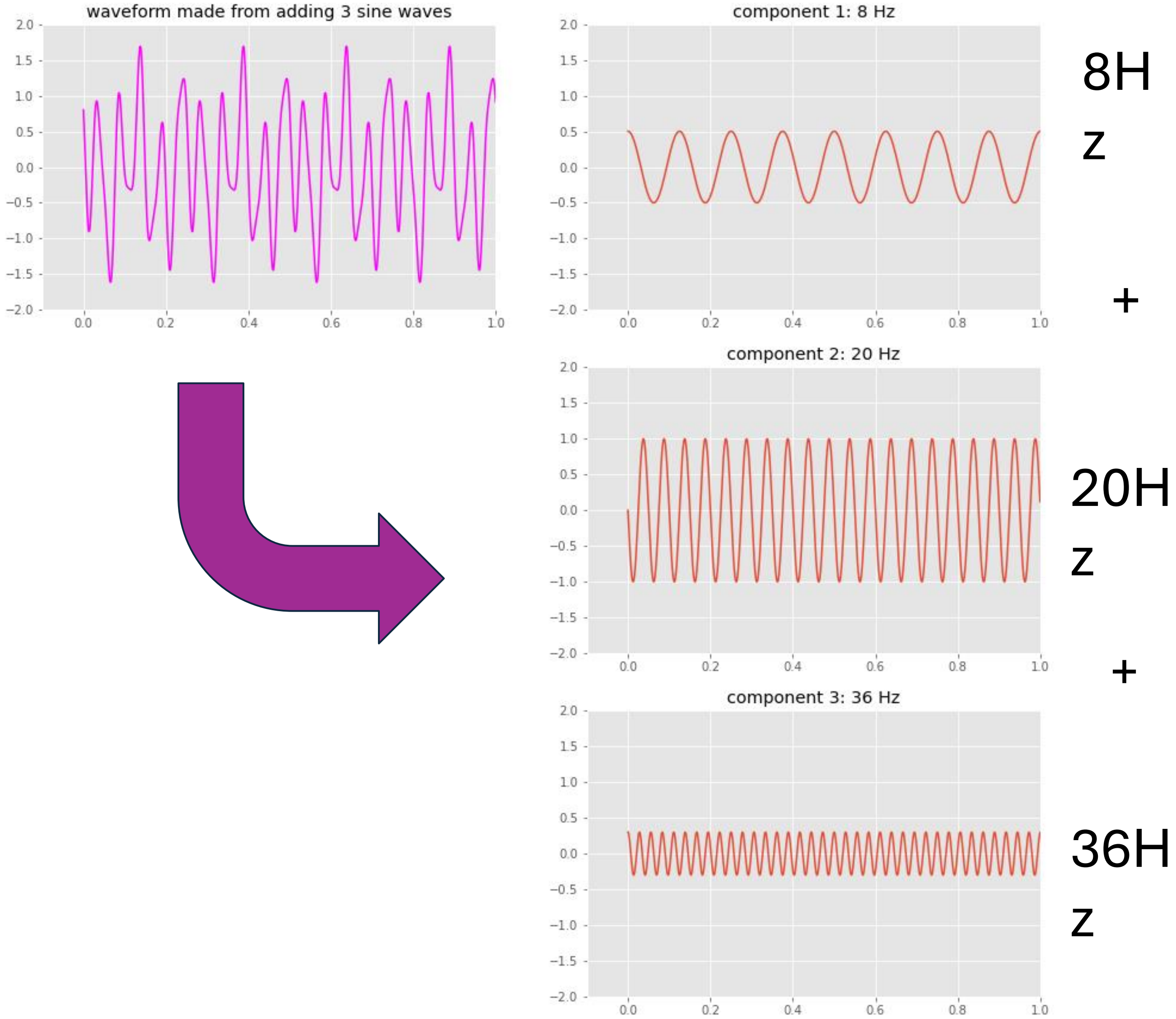






Fourier Transform

If we **scale** and **shift** those pure tones appropriately we can approximate the original waveform by adding the scaled and shifted waves together



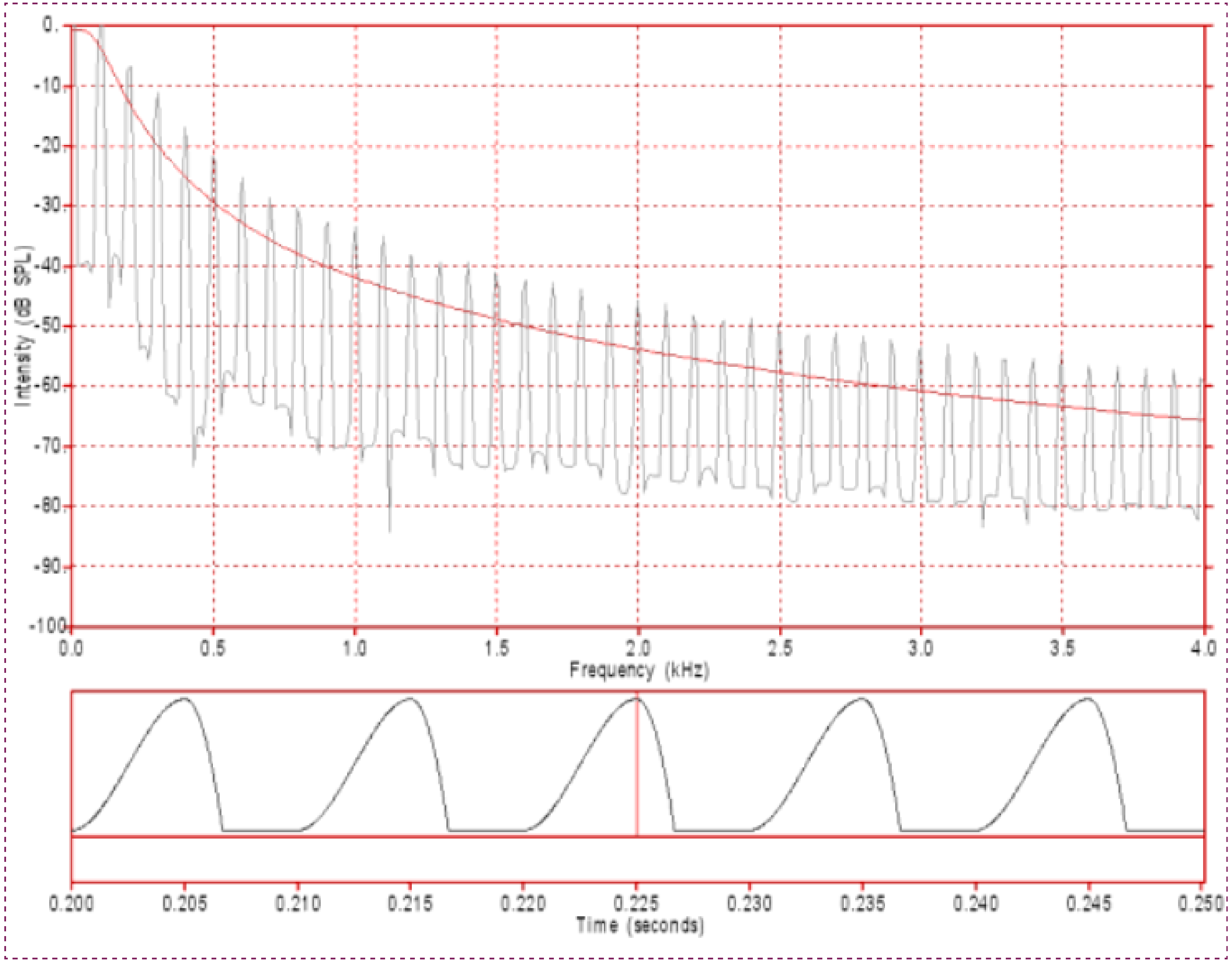
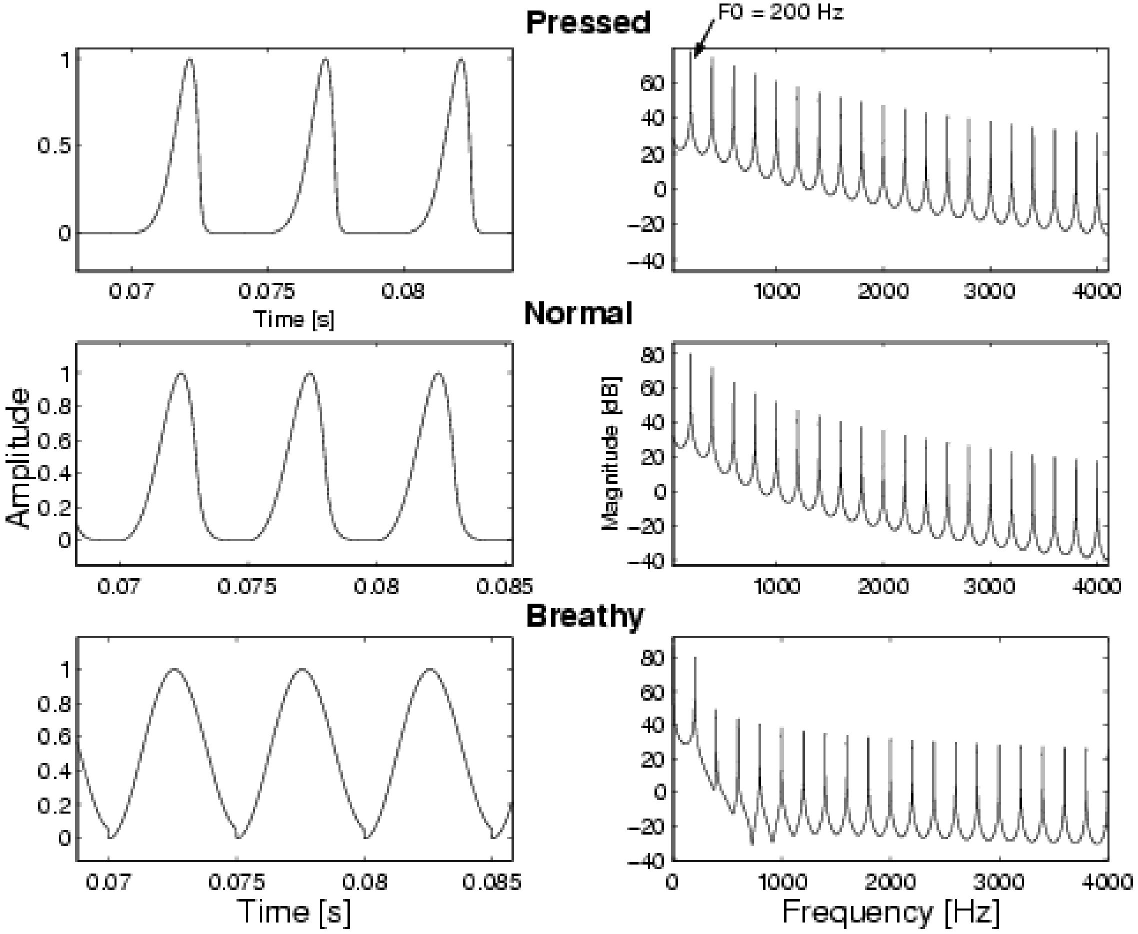


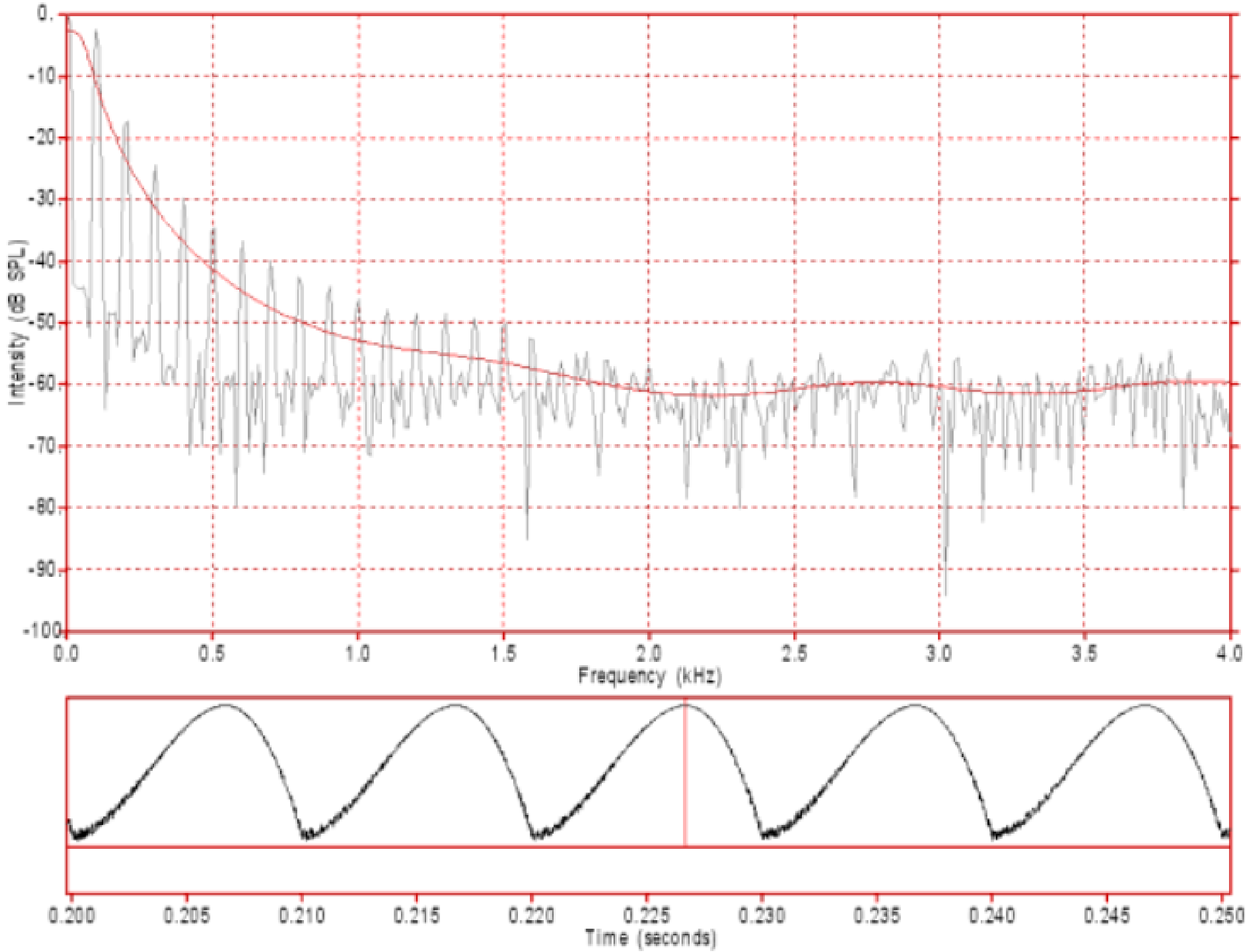
Figure 2: The waveform (bottom) and spectrum (top) of a periodic series of synthesised glottal pulses simulating modal phonation. Each glottal pulse is identical to the pulse in figure 1 and the glottal pulses occur at the rate of 100 per second. Each glottal cycle has an open phase of 6.7 ms and a closed phase of 3.3 ms for a total glottal cycle of 10 ms. Another way of expressing this is to refer to this as an open quotient of 0.67 (ie. 67% of the cycle is open). **Click on the image to hear the sound.**

Fundamental frequency

• Average F0 values for male, female, and children are about 120, 220 and 330 Hz

• The shape of the glottal pulse is individual, and it is an important determinant of the perceived voice quality ("She has a harsh / rough / breathy / clear /.. voice")

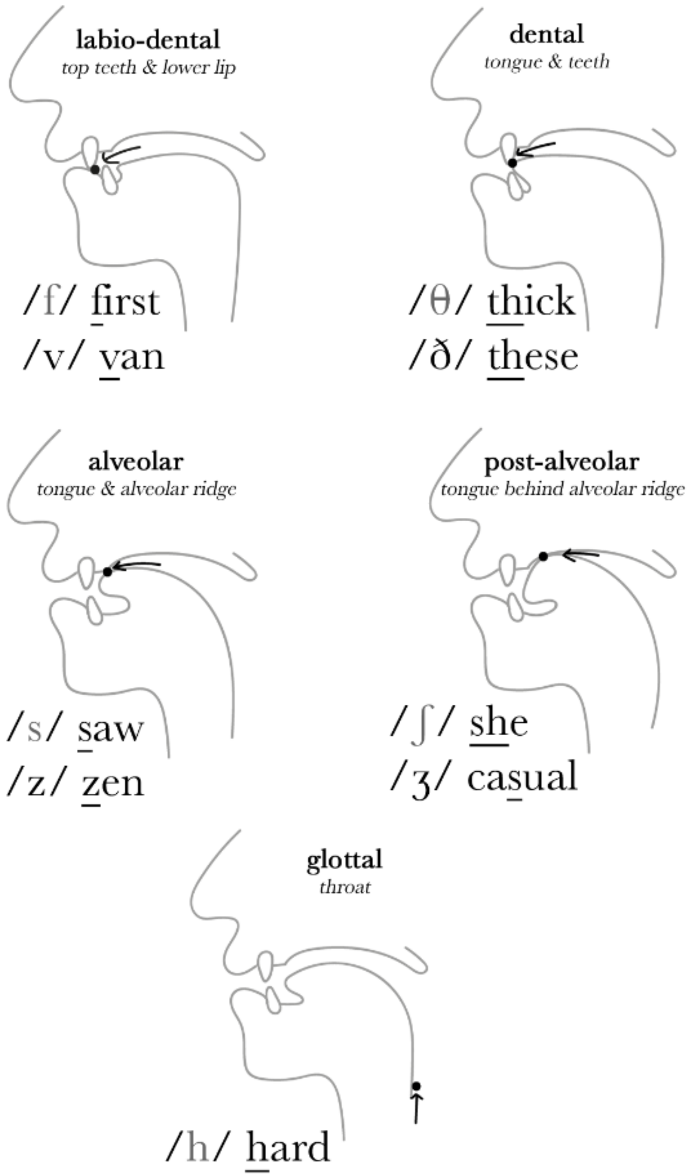




حرف ال ح

Figure 4: The waveform (bottom) and spectrum (top) of a periodic series of synthesised glottal pulses simulating a very breathy voice. Note the noise in both the waveform (particularly in the dips) and the spectrum (particularly above 1.6 kHz). The glottal pulses occur at the rate of 100 per second and each glottal cycle has a period of 10 ms (0.01 s). These glottal pulses have a rise (opening phase) time of about 6.67 ms and a fall (closing phase) time of about 3.33 ms (opening to closing ratio of about 2:1). Each glottal cycle has an open phase of 10 ms and a closed phase of 0 ms for a total glottal cycle of 10 ms. That is, it has an open quotient of 1.0 (100%). **Click on the image to hear the sound.**

Fricative Consonant Sounds



The fricative sounds /v,ð,z,ʒ/ are voiced, they are pronounced with vibration in the vocal cords, whilst the sounds /f,θ,s,ʃ,h/ are voiceless; produced only with air.

Aperiodic Sound Source Spectra (voiceless fricatives)

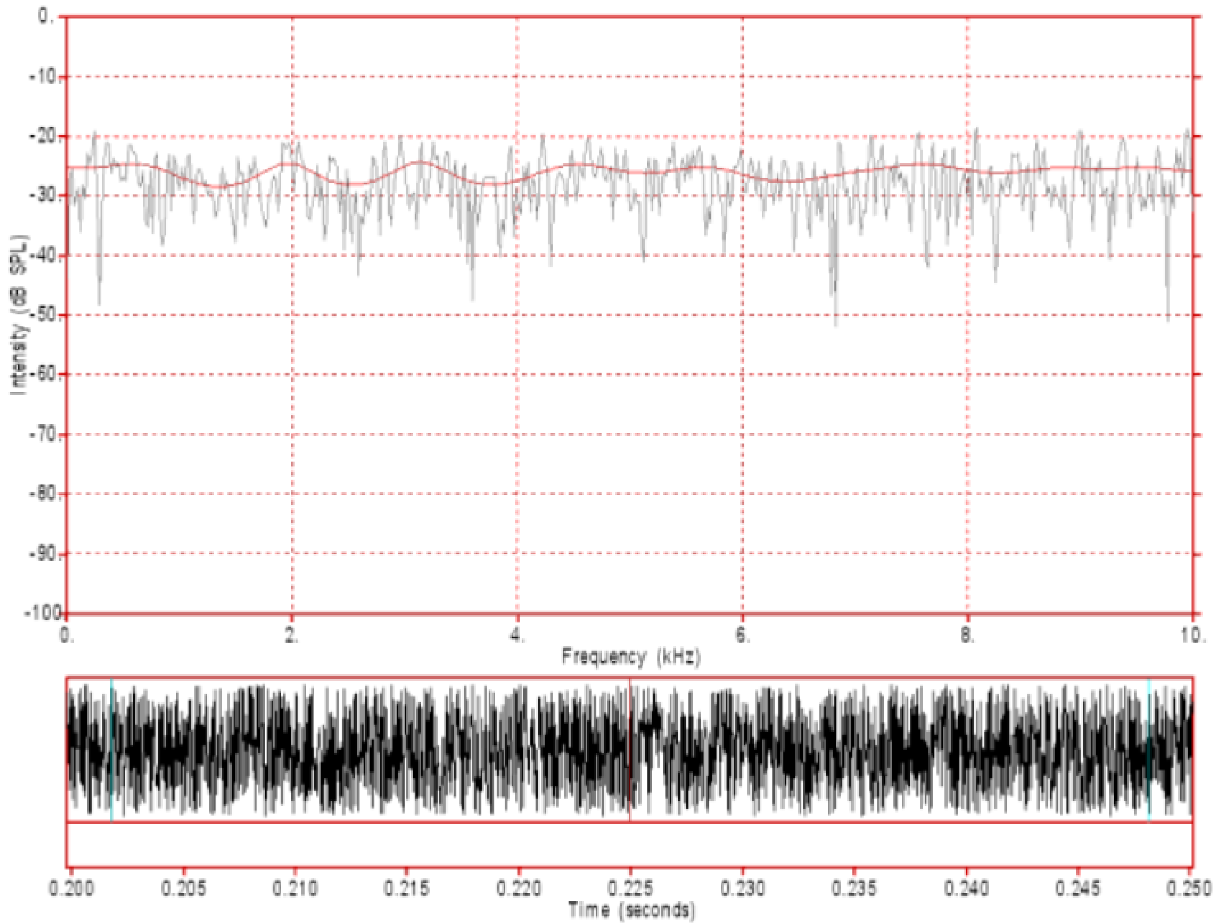
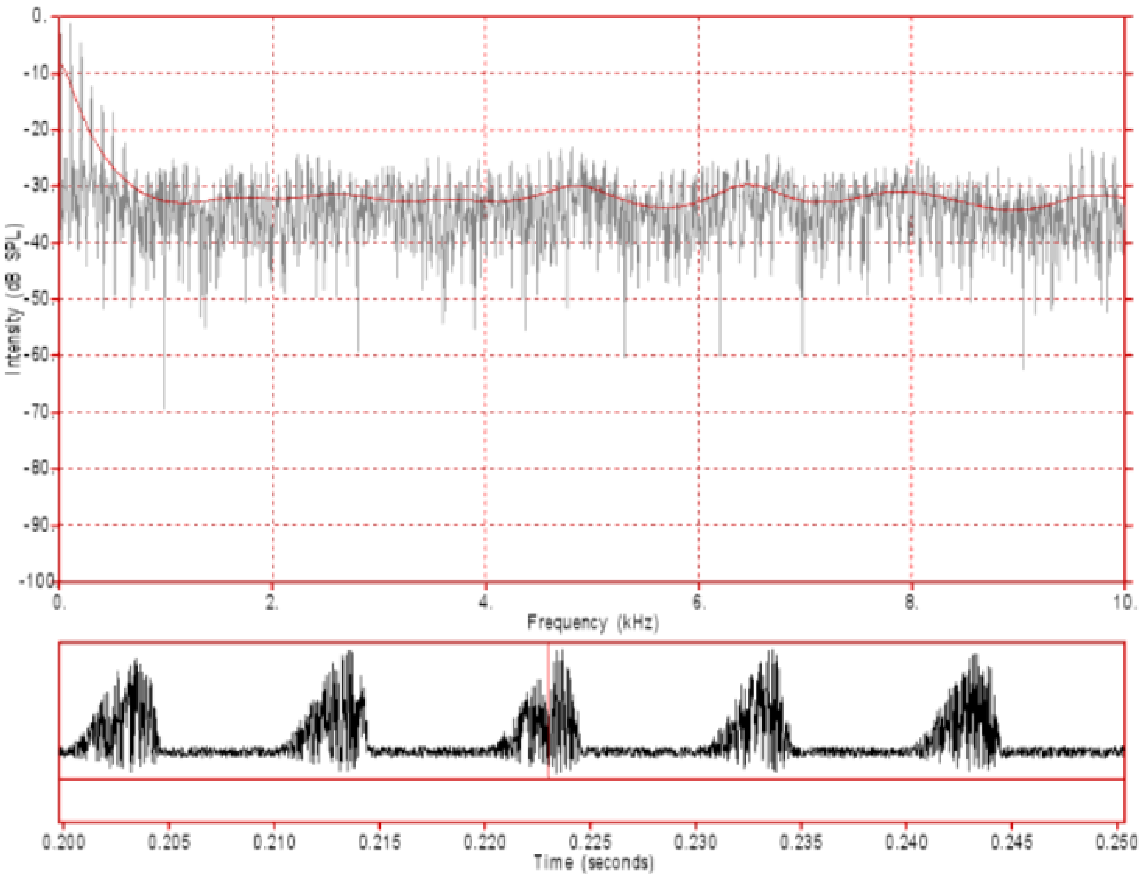


Figure 10: White noise used as a simplified model of a fricative sound source. Note the random pattern of both the waveform (bottom) and the spectrum (top). Also note that the spectral envelope (LPC spectrum in red) is approximately flat. **Click on the image to hear the sound.**

Voiced Fricative Sound Sources



Letter v

Figure 13: This figure displays a composite source that combines the low frequency component of a voiced source plus a noise source modulated by the voiced source. Note, in particular, the strong harmonics below 1000 Hz as well as the weaker, modulation generated, harmonics mixed with noise at higher frequencies. **Click on the image to hear the sound.** Hear a differentiated version of this noise source.